SOFTWARE ENGINEERING LAB 1

Problem Statement: Incident Escalation & On-Call Rotation Engine

Name: Shreya Shenoy
SRN: PES2UG24CS487
Section: 5H

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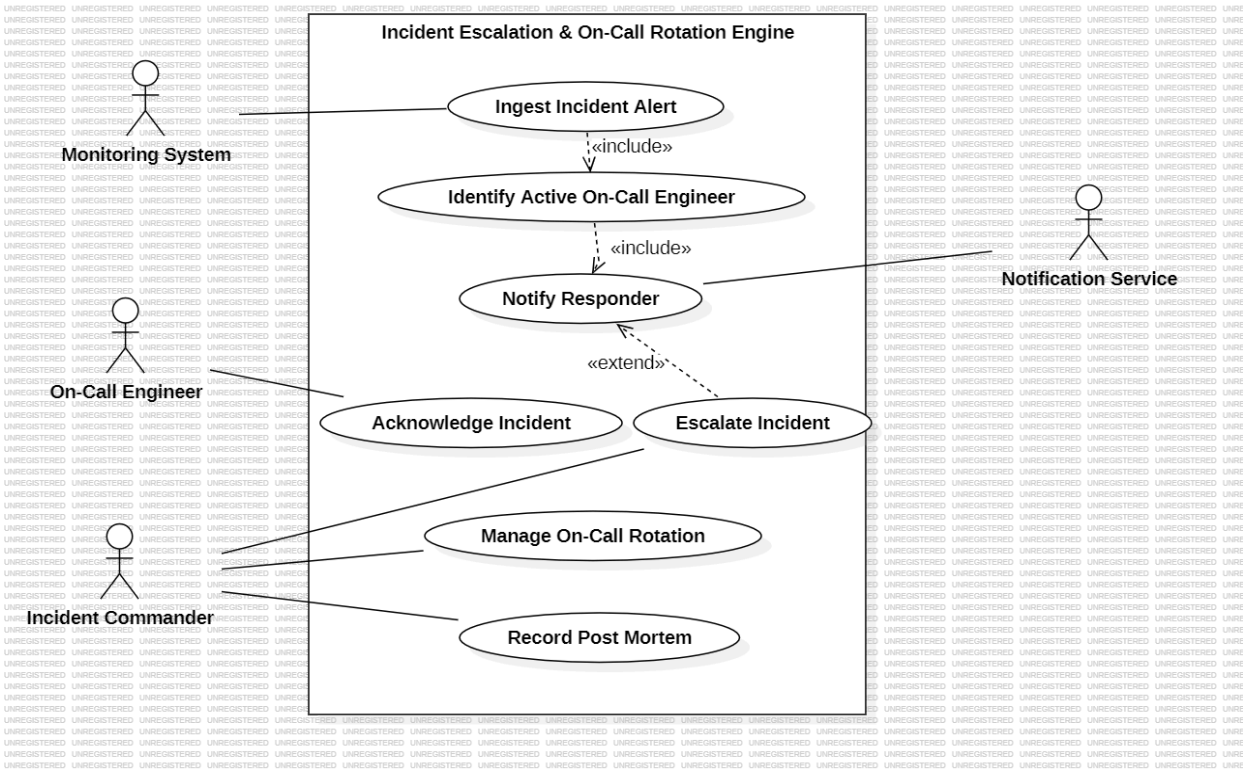
Deliverable 1: Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall ingest incident alerts, identify the active on-call engineer from the rotation schedule, and escalate to the secondary engineer if unacknowledged within 5 minutes.	High	Pass if an unacknowledged incident is escalated to the secondary engineer after 5 minutes. Fail if the incident remains un-escalated after the timeout.	Ensures critical incidents are not missed by the primary on-call engineer.	Timeout is clearly specified as 5 minutes; escalation target should be configurable if multiple tiers are supported.
FR-002	Functional	The system shall allow the Incident Commander to create and update on-call rotation schedules.	High	Pass if a valid rotation schedule is successfully created or updated and the assigned engineer is displayed for the specified shift. Fail if the schedule is not saved or the assigned engineer is incorrect.	Ensures incidents are routed to the correct engineer during each on-call shift.	Specify whether only the Incident Commander can modify schedules and whether overlapping shifts are permitted.
FR-003	Functional	The system shall notify the active on-call	High	Pass if a	Enables the	Specify the

		engineer when a new incident alert is received.		notification is initiated for the active on-call engineer after a valid incident alert is received. Fail if no notification is initiated.	assigned engineer to become aware of incidents and respond promptly	notification channels to be used, such as SMS, phone, email, or webhook.
FR-004	Functional	The system shall escalate an unacknowledged incident to the next responder in the configured escalation hierarchy.	High	Pass if the next responder is notified after the configured acknowledgement timeout expires. Fail if escalation does not occur after the timeout.	Prevents critical incidents from remaining unattended.	Specify the timeout duration and maximum number of escalation levels.
FR-005	Functional	The system shall allow the Incident Commander to record and track a post-mortem for a resolved incident.	Medium	Pass if a post-mortem can be created, associated with the resolved incident, and retrieved later. Fail if it cannot be saved or retrieved	Supports incident analysis and future incident prevention.	Specify the mandatory information required in a post-mortem, such as root cause, impact, timeline, and corrective actions.
NFR-001	Nonfunctional(Performance & Security)	The system shall initiate alert dispatch via webhook, SMS, and email within 3 seconds of alert ingestion.	High	Pass if alert dispatch is initiated within 3 seconds under simulated peak load. Fail if dispatch exceeds 3 seconds.	Ensures rapid notification during critical incidents.	Verify the 3-second target under peak load and specify the security requirements for each notification channel.
NFR-002	Nonfunctional (Reliability)	The system shall preserve incident, acknowledgement, escalation, and on-call schedule records without data loss during normal operation.	High	Pass if all records remain retrievable after simulated incident operations. Fail if any stored record is lost or	Reliable records are necessary for escalation tracking and post-morte	Specify the required data retention period and recovery expectations in case of system failure.

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Deliverable -2 :UML Use-Case Diagram



Deliverable-3 : Use-Case Flow Document

Use Case	**Goal**	**Actors**	**Main Flow**	**Alternate Flow**
Escalate Incident	To ensure that an unacknowledged incident is successfully routed to the next available responder.	On-Call Engineer, Notification Service, Incident Commander	**1.** System detects that the active On-Call Engineer has not acknowledged the incident within 5 minutes. **2.** System retrieves the next responder from the escalation hierarchy. **3.** System sends an escalation	**1.** Next responder does not acknowledge the incident within the configured timeout. **2.** System detects the timeout. **3.** System escalates the incident to the Incident Commander. **4.** System sends an escalation

			notification to the next responder. **4.** Next responder receives the notification. **5.** Next responder acknowledges the incident. **6.** System records the acknowledgement. **7.** System updates the incident status to Acknowledged.	notification to the Incident Commander. **5.** Incident Commander acknowledges or takes control of the incident. **6.** System records the escalation and updates the incident status.